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Digital Product Passport - access rights management, information system security, and business confidentiality

Passeport numérique des produits - Gestion des droits d'accès, sécurité du système d'information et confidentialité des affaires

Digitaler Produktpass - Management der Benutzerrechte, IT-Sicherheit und Geschäftsgeheimnisse

This draft European Standard is submitted to CEN members for enquiry. It has been drawn up by the Technical Committee CEN/CLC/JTC 24.

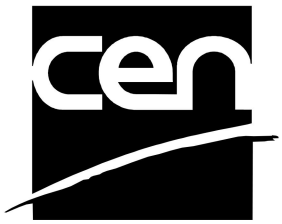
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European foreword

This document (prEN 18239:2025) has been prepared by Technical Committee CEN/CLC JTC 24 “Digital Product Passport - Framework and System”, the secretariat of which is held by DIN.

This document is currently submitted to the CEN Enquiry.

This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

Introduction

This document aims at harmonizing the identity management that ensures that organisations, individuals, machines and services are provided with acknowledged identities. This document defines clear rules and requirements related to access control measures to regulate the access to restricted product passport information.

This document defines rules and requirements related to:

— access right management;

— access control;

— exchange of access right information between economic operators, back-up system operators and registry of the European Commission;

— measures to regulate the access to restricted product passport information;

— possibility for product group specific definition of access rights by delegated acts;

— requirement on information system security;

— requirements on business confidentiality and their representation in access rights management;

— differentiate access rights of different user groups and authorities;

— rules to guarantee IT-security, cybersecurity, and data protection; and

— mechanism on how to transfer responsibilities, access-rights, and data from one economic operator to another.

EXAMPLE When a DPP will need to be updated to include information related to repair activities performed by a professional repairer.

1 Scope

This document specifies the requirements for Digital Product Passport (DPP) access rights management, including IT security, data protection, and responsibility transfer between economic operators. It defines the framework for managing confidential information access, while acknowledging that public DPP data requires no access restrictions. The document applies to all product groups subject to DPP requirements under Regulation (EU) 2024/1781, with specific access rights to be detailed in respective delegated acts.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN ISO 22301:2019[1], *Security and resilience — Business continuity management systems — Requirements (ISO 22301:2019)*

EN ISO/IEC 27000:2020, *Information technology — Security techniques — Information security management systems — Overview and vocabulary (ISO/IEC 27000:2018)*

EN ISO/IEC 27001:2023, *Information security, cybersecurity and privacy protection — Information security management systems — Requirements (ISO/IEC 27001:2022)*

ISO/IEC 27031:2025, *Cybersecurity — Information and communication technology readiness for business continuity*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply:

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp

— IEC Electropedia: available at https://www.electropedia.org/

3.1
controlled DPP data
information on digital product passport whose access is controlled based on the user's access rights

Note 1 to entry: User: person who interacts with a system, product or service [SOURCE: ISO 26800:2011, 2.10; modified, Notes changed]

3.2
system resilience
ability to recover from security compromises or attacks

[SOURCE: ISO/IEC 29180:2012]

3.3
system availability
property of being accessible and usable on demand by an authorized entity

[1] As impacted by EN ISO 22301:2019/A1:2024.

[SOURCE: EN ISO/IEC 27000:2020]

3.4
cyber resilience
ability to maintain business continuity despite adverse conditions, attacks, or compromises on critical data flow and related information systems

3.5
RPO
recovery point objective
point in time to which data must be recovered after a disruption has occurred

[SOURCE: ISO/IEC 27031:2025, 3.12]

3.6
RTO
Recovery Time Objective
period of time within which minimum levels of services and/or products and the supporting systems, applications, or functions must be recovered after a disruption has occurred

[SOURCE: ISO/IEC 27031:2025]

3.7
actor
organization or individual that fulfils a role

[SOURCE: ISO 23234:2021, 3.4]

3.8
notified actor
organization or individual entitled by an authorized accrediting body or authority, that fulfils a role in the DPP lifecycle

3.9
digital product passport
DPP
digital record of product characteristics throughout its life cycle

Note 1 to entry: Example characteristics include environment sustainability, environmental impact and recyclability

4 Business transactions and responsibilities related to the DPP

4.1 Business aspects of the DPP lifecycle as basis of access management

Based and informed by EN ISO 11354-1:2011, understanding product lifecycle phases and stages, stakeholders along the product life cycle, as well as relevant business transactions occurring in this lifecycle, is indispensable to defining the relevant business responsibility and access management requirements along the lifecycle stages of Digital Product Passports. The lifecycle phases and stages are outlined in Annex A.

In general, not all the phases and the stages mentioned in this document might be applicable to every product sector. Therefore, only the phases and stages relevant for a specific product group shall be

considered. Moreover, in case a product group specific stage is not represented in the following list, it should be added where relevant.

4.2 Stakeholders along the DPP lifecycle

This section identifies the main stakeholders who are responsible for the handling of a DPP relevant product across the value chain and therefore need to access a DPP along its business stages. The identified stakeholders are the following:

1. **Economic operators (in short EO)**: a top-level role encompassing the manufacturer, the authorized representative, the importer, the distributor, the dealer and the fulfilment service provider.

    a. **Manufacturers**: any natural or legal person that manufactures a product or that has a product designed or manufactured and markets that product under their name or trademark.

    b. **Authorized representatives** (of non-EU companies in the EU market): any natural or legal person established in the Union that has received a written mandate from the manufacturer to act on the manufacturer's behalf in relation to specified tasks with regard to the manufacturer's obligations.

    c. **Importers**: any natural or legal person established in the Union that places a product from a third country on the Union market.

    d. **Distributors**: any natural or legal person in the supply chain, other than the manufacturer or the importer, that makes a product available on the market.

    e. **Dealers**: a distributor or any other natural or legal person that offers products for sale, hire or hire purchase, or that displays products, to end users in the course of a commercial activity, including through distance selling; and includes any natural or legal person that puts a product into service in the course of a commercial activity.

    f. **Fulfilment service providers**: any natural or legal person offering, in the course of commercial activity, at least two of the following services: warehousing, packaging, addressing and dispatching, without having ownership of the products involved, excluding postal services, parcel delivery services, and any other postal services or freight transport services.

2. **Other value chain actors**

    a. **Customers**: a natural or legal person that purchases, hires or receives a product for their own use whether or not acting for purposes which are outside their trade, business, craft or profession.

    b. **Professional repairers**: a natural or legal person that provides professional repair or maintenance services for a product, irrespective of whether that person acts within the manufacturer's distribution system or independently.

    c. **Independent operators**: means a natural or legal person that is independent of the manufacturer and is directly or indirectly involved in the refurbishment, repair, maintenance or repurposing of a product, and includes waste management operators, refurbishers, repairers, manufacturers or distributors of repair equipment, tools or spare parts, as well as publishers of technical information, operators offering inspection and testing services and operators offering training for installers, manufacturers and repairers of equipment.

    d. **Recycler**: any natural or legal person who is responsible for the collection, processing and recovery of reusable materials and who applies a decontamination process.

e. **Market surveillance authority**: an authority designated by a Member State as responsible for carrying out market surveillance in the territory of that Member State.

3. **Customs authorities:** customs authorities as defined in point 1 of Article 5 of Regulation (EU) No 952/2013.

4. **Service provider**: any natural or legal person providing an information service.

   a. **Digital product passport service provider** (main or back-up): a natural or legal person that is an independent third-party authorized by the economic operator which places the product on the market or puts it into service and that processes the digital product passport data for that product for the purpose of making such data available to economic operators and other relevant actors with a right to access those data.

      1. **Back-up digital product passport service provider** as "digital product passport service provider that is hosting the back-up copy of the digital product passport" [prEN 18221].

5. **Parties not defined by the ESPR**: Such as for example system administrators or non-EU state bodies.

5 Functional Requirements for business confidentiality and access rights requirements

5.1 General

Based on the product lifecycle as per Annex A and the different business roles along this lifecycle [Clause 4.2] the following axioms for DPP access rights management, information system security and business confidentiality can be distilled.

5.2 Functional requirements for business confidentiality

1. Actors in a DPP system need to have globally unique operator identifiers, as defined in prEN 18219.

2. Access to public data shall be possible for EU and non-EU actors without additional authentication.

3. Access to DPP controlled data and its management shall be possible for EU and non-EU actors.

4. Obtaining a globally unique operator identifier shall be possible outside of the EU for non-EU actors.

5. Authentication of actors accessing or managing controlled DPP data shall allow for accountability and be non-refutable.

6. It shall be possible to verify the authenticity of a globally unique operator identifier ("trust").

7. Each Economic Operator may assign, based on a globally unique operator identifier, certain rights and roles to any notified actor that wishes to access its Digital Product Passport controlled data provided that the requesting notified actor fulfils the legal requirements.

8. Base roles (for instance "recycler" or "refurbisher") which grant access to DPP's controlled data will be defined by the delegated acts. The Economic Operators can however create additional roles.

9. Globally unique operator identifiers of notified actors fulfilling state or official base roles need to be attached to a system, which allows them to carry trustworthy credentials, allowing for access to controlled data without the need of individual Economic Operator's explicit approval.

10. For actors fulfilling additional roles not regulated but defined by the Economic Operator, the Economic Operator can decide which authentication mechanism to apply.

11. As roles can evolve over time, access rights shall be defined at property level and defined through controlled vocabularies.

12. Search functionalities in EU registries and DPP services shall be limited in such a way as to prevent mass data scraping. Technical solutions providing exceptions for actors fulfilling state or regulated base roles shall be provided where necessary.

13. To counteract DPP counterfeit, where authentication of the DPP issuer is required, the Economic operator shall be securely identified.

14. When authentication is required to access any service related to DPP management (e.g. issuing, amendment and access to controlled data), such authentication shall be performed with a Level of Assurance at least equivalent to the high level as required by the relevant legal act.

15. The re-use of public data cannot be avoided. It is recommended, where needed, to take necessary measures to spot, avoid or detect anomalies.

16. An Economic Operator can introduce “access rate limiting” in case of reasonable, identified risks or legal requirement, if legal information requirements are not affected.

17. There shall be no additional burden on the Economic Operator’s in terms of software deployment, with respect to the obligation already foreseen by its responsibilities.

18. All access and changes to roles, rights, or data shall be logged and auditable.

19. Any access rights management system managing access to controlled DPP data shall have clear and documented processes for granting, revoking or modifying access rights when roles change, or actors are no longer authorized.

20. There shall be the possibility to delegate access to roles. The actor receiving the delegation shall fulfil the legal obligation the role requires.

21. A mechanism to handle emergency revocations of access rights in case of breach or non-compliance is necessary.

22. The DPP Framework and system shall provide a mechanism to define role-based access control. The DPP system shall describe how the root of trust is established that can issue such roles.

23. Access management security shall be evaluated by all relevant actors concerned as per EN ISO/IEC 27000:2020 and EN ISO/IEC 27001:2023.

24. Access to all controlled DPP data shall be authenticated and authorized before being granted. Authentication shall be performed with a Level of Assurance at least equivalent to the high level as required by the relevant legal act. Least privilege principles shall be implemented.

25. Access rights shall be defined per each controlled data element.

6 Requirements on system and service resilience

6.1 General requirements

There shall be no authentication to access public data. The authentication functionality shall be simple, reliable and secured. To access and amend controlled data, the authentication is required.

Authentication includes the authentication process for users and applications. Each authenticating process requires data transmission, reception, and shall be easy to use.

6.2 Access management

From the functional requirements enunciated in Clause 5 the following requirements for access management arise.

Economic Operators control the access to DPP data of products they are liable for. Access schemes to DPP data can be separated into four categories:

1. **Access by authorities:** these actors have read only access to a DPP. However, authorities have access to restricted information (controlled data) and hence shall be authenticated against an acknowledged certified scheme.

2. **Access to public data:** any actor has unauthenticated read-only access to publicly available DPP information.

3. **Access by other parties acting in a role defined by an implementing or delegated act,** e.g. role 'refurbisher': such actors have read-write access to controlled data of a DPP. Therefore, an actor, which performs read/write operations shall be identified unambiguously against an acknowledged, regulated certified scheme and providing a level of assurance at least equal to “High” as defined in the relevant legal act.

4. **Access granted by an Economic Operator issuing a DPP to arbitrary non-regulated actors** (e.g. not defined in the scope of the regulations/delegated act): such actors have read-write access to controlled data. In such scenarios authentication shall provide a level of assurance at least equal to “High” as defined in the relevant legal act.

5. **Access by a consumer to its controlled data** (e.g. data deriving from its usage of the product): consumers may have read only access to their data. Therefore, in such a scenario a consumer shall be identified unambiguously against an acknowledged certified/regulated scheme.

6.3 Access revoking mechanism

Following the requirement in Clause 5.2.21 a policy for “access revocation” shall be put in place in all Identity Management Systems (IMS) in use in regulated DPP related services.

6.4 Digital operational resilience

6.4.1 General

Digital operational resilience aims at the resilience of the overall supply chain of ICT services [references: ISO/IEC 27036-3:2023, ISO 28003:2007], ICT systems or ICT products throughout their lifecycle against disruptive events.

6.4.2 Business continuity

Business Continuity Management (BCM) reduces risks from attacks against the availability of systems and shall be implemented according to the following international standards:

— EN ISO 22301:2019, Security and resilience - Business continuity management systems – Requirements.

— ISO/IEC 27031:2025, Information technology - Security techniques - Guidelines for information and communication technology readiness for business continuity.

NOTE Attention is drawn to existing local regulations that can apply.

6.4.3 Continuous improvement

This clause outlines the continuous improvement process for Information Security Management System (ISMS) in accordance with EN ISO/IEC 27001:2023. The process for establishing and maintaining ISMS shall be based on a Plan-Do-Check-Act (PDCA) cycle to ensure systematic enhancement of information security.

— **Planning (Plan)** The organization shall establish objectives and scope for the ISMS. A comprehensive risk assessment shall be conducted to identify potential threats and vulnerabilities. Appropriate security controls shall be selected and documented based on the risk assessment.

— **Implementation (Do)** The organization shall implement the security controls defined in the planning phase. All personnel shall be trained and made aware of their roles and responsibilities regarding information security. Necessary security measures shall be deployed, and policies shall be enforced.

— **Monitoring and review (Check)** The effectiveness of the implemented controls shall be continuously monitored and reviewed. Key performance indicators (KPIs) shall be measured and analysed to assess the health of the ISMS.

— **Improvement (Act)** Corrective and preventive actions shall be taken based on the results of monitoring and review. Processes shall be proactively enhanced to improve the ISMS. Continuous improvement shall be ensured by addressing identified issues and implementing necessary changes.

6.5 Security management

6.5.1 Service availability

The availability and timely accessibility of digital product data are essential as defined by/in implementing acts.

6.5.2 Security by design

Critical data flows and related information systems shall be implemented as security by design and by default. Measures on how to uphold and preserve reliable, resilient and secure information systems during the whole lifecycle of critical DPP data shall be identified and implemented. Thereby the following aspects shall be assessed and considered:

— overall quality and resilience of products and services;

— cybersecurity risk-management measures embedded in them, and

— cybersecurity practices of their suppliers and service providers, including their secure development procedures.

The inclusion of cybersecurity risk-management measures into contractual arrangements with direct suppliers and service providers shall be encouraged.

6.5.3 Detect and response

Cybersecurity risk-management measures should consider critical data flows and related information systems and include measures related to the prevention, detection, identification, containment, analysis and response to incidents.

6.5.4 Recovery capability

Critical data flows and related information systems shall be equipped with redundant systems and back-up working space to ensure continuity. There shall be resilience mechanisms implemented to mitigate against Denial-of-Service (DoS) attacks and allow for the recovery to a stable state if such an attack happens.

Annex A
(informative)

Basis for business phases

The digital product passport's aim to *accompany* products relies on its capability to follow the product through all the phases and stages of its lifecycle. To establish effective access management, these phases and stages shall be identified, understood and consolidated within a structured framework.

To capture all the relevant access management aspects of a product's lifecycle the following *high-level* phases and the corresponding stages, which can occur in each phase, are identified, primarily instructed by the ESPR.

In general, not all the phases and the stages mentioned in this document might be applicable to every product sector. Therefore, only the phases and stages relevant for a specific product group shall be considered. Moreover, in case a product group specific stage is not represented in the following list, it should be added where relevant.

1. **Concept phase:** period of time of a product's life cycle during which the user needs are identified and the product's concepts are described and evaluated [Source: ISO/IEC/IEEE 24765:2017, 3.749]

2. **Design phase:** period during which the engineering representation of a product is dynamic [Source: ISO 10303-2:2024, 3.1.274]

3. **Procurement phase of components:** phase, in which are executed processes which create, manage and fulfil contracts relating to the provision of goods, services and construction works or disposals, or any combination thereof [Source: ISO 6707-2:2017, 3.5.18].

NOTE In this phase the DPP data are to be obtained by the manufacturer from its suppliers.

In this phase can occur:

a. **Acquisition**: process of obtaining a product, service or system [source: ISO/IEC/IEEE 24748-7000:2022, 3.1.2].

NOTE As example, procurement of intermediate products, as for instance aluminium, steel, consumables, sub-systems, parts, etc.

b. **Generation from natural resources**: mentioned in the relevant regulatory act. [ESPR Art. 2 (12)]

Note: as example, the wood harvesting by the wood processing and/or wooden products manufacturing plant.

4. **Production phase**: production, manufacturing and packaging operations [Source: ISO 22716:2007, 2.26].

    a. **Pre-Processing**: mentioned in the relevant regulatory act. [ESPR Art. 2 (12)]

    b. **Manufacturing:** process and action performed by an equipment supplier/manufacturer that are necessary to provide finished component(s), assembly(ies) and related documentation, that fulfil the requests of the user/purchaser and meet the standards of the supplier/manufacturer.

5. **Qualification phase:** proof of suitability of an individual, process, procedure or service to fulfil specified requirements [Source: ISO 16528-1:2007, 2.13].

6. **Operational phase:** product life cycle phase comprising all technical, administrative and managerial actions, other than maintenance actions, that result in the supported product functioning according to its intended use [Source: ISO 19650-3]. A manufactured product is placed on the market in the operational phase via an agreement or offer to change ownership, wherefore the obligation to have a DPP starts. In this phase the following stages can occur:

    a. **Storage**: life cycle phase of a physical product being kept available in an adequate environment [Source: ISO/TS 22002-5:2019, 3.14].

    b. **Distribution of goods:** movement of products from their point of origin to their destination and consisting of the following basic elements: packaging, unit loads, material-handling systems, storage facilities and means of transportation [Source: ISO 3676:2012, 3.1].

    c. **Installation**: product life cycle phase in which a product is prepared such that it fulfils its intended use [Source: ISO 24183:2024, 3.3.4.1].

    d. **Use:** The phase, in which the product's function is used or maintained and value created.

    NOTE It also includes repair, redesign, and modification activities of a Product at the servicing location] [Source: ISO 10303-232:2002, F.1.202].

7. **Maintenance phase:** involves a range of activities to prevent malfunctions, failures, and hazards. It encompasses preventive and corrective measures to ensure the reliable operation of technical systems [Source: DIN-31051]. These activities include:

    1. **Maintaining**: maintenance measures aim to maintain the target condition of technical systems. This involves standardizing maintenance processes, implementing relevant management measures, and ensuring that machines and components operate within their intended parameters. [Source: DIN-31051]

    2. **Repairing**: inevitably, machines and equipment may experience defects or failures. Maintenance standards emphasize the importance of repair measures to restore functionality. By addressing faulty components and eliminating malfunctions, maintenance professionals can extend the lifespan of machines and prevent further damage. [Source DIN-31051]

    3. **Improving/ Upgrading**: maintenance is not solely about restoring functionality; it also involves continuous improvement. Maintenance standards encourage maintenance professionals to analyse failures, identify potential areas for improvement, and implement measures to enhance the efficiency and reliability of machines and equipment. [Source DIN-31051]

    4. **Refurbishment:** renovation and restoration to intended use condition [Source: ISO 10303-232:2002, F.1.202 – also see DIN-31051, 4.1.4]

— **Reuse phase:** any operation by which products or components that are not waste are used again for the same purpose for which they were conceived [Source: Directive 2008/98/EC]. In this phase the following stages can occur:

    1. **Preparing for re-use**: checking, cleaning or repairing recovery operations, by which products or components of products that have become waste are prepared so that they can be re-used without any other pre-processing.

    2. **Re-use**: any operation by which products or components that are not waste are used again for the same purpose for which they were conceived.

3. **Recycle:** means any recovery operation by which waste materials are reprocessed into products, materials or substances whether for the original or other purposes. It includes the reprocessing of organic material but does not include energy recovery and the reprocessing into materials that are to be used as fuels or for backfilling operations.

4. **Recovery:** means any operation the principal result of which is waste serving a useful purpose by replacing other materials which would otherwise have been used to fulfil a particular function, or waste being prepared to fulfil that function, in the plant or in the wider economy.

8. **End-of-life phase:** the life cycle phase that begins when a product is discarded and ends when the waste material of the product is returned to nature or enters another product's life cycle; [Source: ESPR Art. 2 (16)]. In this phase the following stages can occur:

    1. **Disassembly:** taking apart of an assembled product into constituent materials and/or components.

    2. **Disposal**: means any operation which is not recovery even where the operation has as a secondary consequence the reclamation of substances or energy.

Table A.1 — High level phases and stages of the product lifecycle.

Phase	Concept[a]	Design[b]	Procurement of components[c]		Production[f]		Qualification [h]	Operational[i]				Maintainance [n]				Re-use[o]				End-of-Life [p]	
Stage	Concept	Design	Acquisition[d]	Generation from natural resources[e]	Pre-processing[e]	Manufacturing[g]	Qualification	Storage[j]	Distribution of goods [k]	Installation [l]	Use [m]	Maintaining [n]	Repairing [n]	Improving/Upgrading [n]	Refurbishment [m, n]	Preparing for reuse[o]	Resuse[o]	Recycle[o]	Recovery[o]	Disassembly [q]	Disposal[o]

[a] Source: ISO-26262.
[b] Source: ISO 10303-2:2024.
[c] Source: ISO 6707-2:2017, 3.5.18.
[d] Source: ISO/IEC/IEEE 24748-7000:2022, 3.1.2
[e] Source: ESPR Art. 2 (12).
[f] Source: EN ISO 22716:2007, 2.26
[g] Source: EN ISO 16070:2005.
[h] Source: ISO 16528-1:2007, 2.13.
[i] Source: EN ISO 19650-3.
[j] Source: ISO/TS 22002-5:2019, 3.14
[k] Source: ISO 3676:2012, 3.1.
[l] Source: ISO 24183:2024, 3.3.4.1.
[m] Source: ISO 10303-232:2002, F.1.202.
[n] Source: DIN-31051.
[n] Source: DIRECTIVE 2008/98/EC.
[p] Source: ESPR Art. 2 (13).
[q] Source: ISO 8887-2:2023, 3.10

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